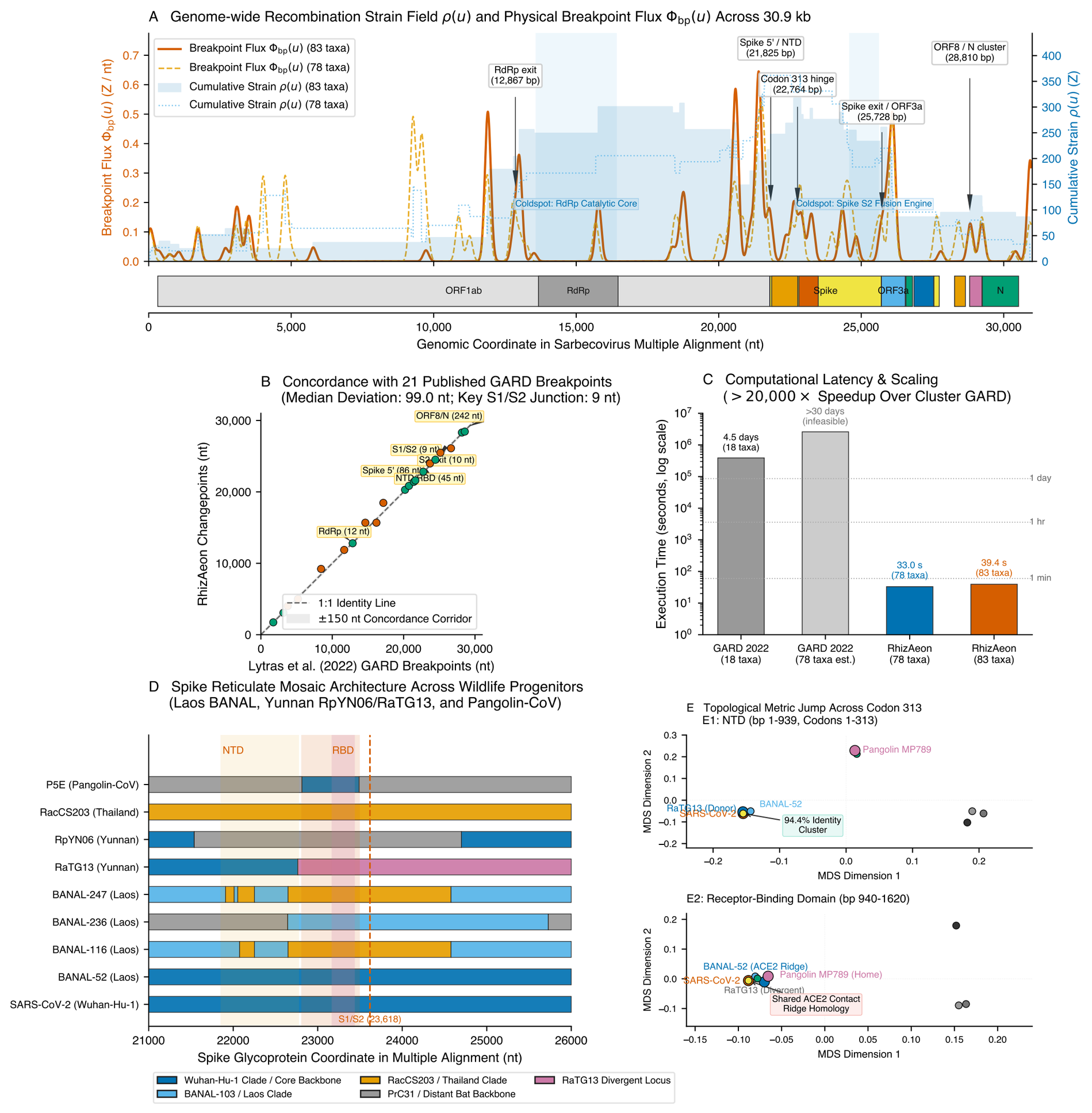

A Genome-wide Recombination Strain Field ρ(u) and Physical Breakpoint Flux Φbp(u) Across 30.9 kb
Breakpoint Flux Φbp(u) (83 taxa)
Breakpoint Flux Φbp(u) (78 taxa)
Cumulative Strain ρ(u) (83 taxa)
Cumulative Strain ρ(u) (78 taxa)
Breakpoint Flux Φbp(u) (Z / nt)
Cumulative Strain ρ(u) (Z)
RdRp exit (12,867 bp)
Spike 5′ / NTD (21,825 bp)
Codon 313 hinge (22,764 bp)
Spike exit / ORF3a (25,728 bp)
ORF8 / N cluster (28,810 bp)
Coldspot: RdRp Catalytic Core
Coldspot: Spike S2 Fusion Engine
ORF1ab
RdRp
Spike
ORF3a
N
Genomic Coordinate in Sarbecovirus Multiple Alignment (nt)
B Concordance with 21 Published GARD Breakpoints
(Median Deviation: 99.0 nt; Key S1/S2 Junction: 9 nt)
ORF8/N (242 nt)
S1/S2 (9 nt)
S2′ exit (10 nt)
Spike 5′ (86 nt)
NTD–RBD (45 nt)
RdRp (12 nt)
1:1 Identity Line
±150 nt Concordance Corridor
RhizAeon Changepoints (nt)
Lytras et al. (2022) GARD Breakpoints (nt)
C Computational Latency & Scaling
(> 20,000 × Speedup Over Cluster GARD)
Execution Time (seconds, log scale)
4.5 days (18 taxa)
>30 days (infeasible)
33.0 s (78 taxa)
39.4 s (83 taxa)
1 day
1 hr
1 min
GARD 2022 (18 taxa)
GARD 2022 (78 taxa est.)
RhizAeon (78 taxa)
RhizAeon (83 taxa)
D Spike Reticulate Mosaic Architecture Across Wildlife Progenitors
(Laos BANAL, Yunnan RpYN06/RaTG13, and Pangolin-CoV)
NTD
RBD
P5E (Pangolin-CoV)
RacCS203 (Thailand)
RpYN06 (Yunnan)
RaTG13 (Yunnan)
BANAL-247 (Laos)
BANAL-236 (Laos)
BANAL-116 (Laos)
BANAL-52 (Laos)
SARS-CoV-2 (Wuhan-Hu-1)
S1/S2 (23,618)
Spike Glycoprotein Coordinate in Multiple Alignment (nt)
Wuhan-Hu-1 Clade / Core Backbone
BANAL-103 / Laos Clade
RacCS203 / Thailand Clade
PrC31 / Distant Bat Backbone
RaTG13 Divergent Locus
E Topological Metric Jump Across Codon 313
E1: NTD (bp 1-939, Codons 1-313)
Pangolin MP789
BANAL-52
RaTG13 (Donor)
SARS-CoV-2
94.4% Identity Cluster
MDS Dimension 1
MDS Dimension 2
E2: Receptor-Binding Domain (bp 940-1620)
BANAL-52 (ACE2 Ridge)
Pangolin MP789 (Home)
SARS-CoV-2
RaTG13 (Divergent)
Shared ACE2 Contact Ridge Homology